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- Subject : M.L

*Assignment No: 05 *

Q4]

⇒ Why it works?

i] Focus on mistakes

- Adaboost trains weak learners, sequentially. After each round, the weights of misclassified samples are increased and correctly classified samples are decreased. The next stump therefore concentrates on the difficult points ignored earlier

ii] weighted voting:

- each stump receives a confidence weight α_i, based on its accuracy. Final prediction is

$$F(x) = \text{sign}\left(\Sigma_t \cdot \alpha_i h_i(x)\right)$$

even if each stump is only slightly better than random, their weighted combination becomes strong because different stumps correct different errors

iii] Margin improvement

Boosting gradually increases the confidence of correct predictions, which improves generalization accuracy

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* When it fails :

- If weak learner can not perform better than perform
- very noisy data or outliers
- low diversity : If all stumps make similar mistakes, ensemble can not improve.

Q2]
⇒

i] Reason 1 : Regularized objective + second-order optimization

- XGBoost uses gradients and Hessions (2nd ~~and~~ order derivation / derivatives) and adds regularization on tree completely

Benefit : more accurate split decisions, reduced overfitting and approximate split finding.

Trade off : More hyperparameters and higher computation per iteration

ii] Reason 2 : Syste

ii] Reason 2: System & algorithm optimizations

- Use sparse-aware learning, parallel processing, subsampling and approximate split finding

Benefits: Faster and reliable on large real world datasets with missing values

Trade off: approximate splits may be slightly less exact and tuning becoming harder, can use more memory

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